

§ 1. The Product Topology and the Tychonoff Theorem

Problem 19.3. *If each space X_α is a Hausdorff space, then $\prod X_\alpha$ is a Hausdorff space in both the box and product topologies.*

Proof. Given $\prod X_\alpha$ with the product topology, let $f, g \in \prod X_\alpha$ be distinct elements. Then there exists some α such that $f(\alpha), g(\alpha) \in X_\alpha$ with $f(\alpha) \neq g(\alpha)$. We can find open sets U_α containing $f(\alpha)$ and V_α containing $g(\alpha)$ such that $U_\alpha \cap V_\alpha = \emptyset$. Quickly note that:

$$\begin{aligned}\pi_\alpha^{-1}(U_\alpha) \cap \pi_\alpha^{-1}(V_\alpha) &= \pi_\alpha^{-1}(U_\alpha \cap V_\alpha) \\ &= \pi_\alpha^{-1}(\emptyset) \\ &= \emptyset.\end{aligned}$$

Both $\pi_\alpha^{-1}(U_\alpha)$ and $\pi_\alpha^{-1}(V_\alpha)$ are open in the product space $\prod X_\alpha$, and likewise $f \in \pi_\alpha^{-1}(U_\alpha)$ and $g \in \pi_\alpha^{-1}(V_\alpha)$. Thus the product space $\prod X_\alpha$ is Hausdorff. Since the box topology is finer than the product topology, $\prod X_\alpha$ is also Hausdorff with the box topology. ■

Problem 19.5. *One of the implications stated in Theorem 19.6 holds for the box topology. Which one?*

Proof. Example 2 which immediately follows Theorem 19.6 gives a counter example for the backwards implication. If $f : X \rightarrow \prod_\alpha X_\alpha$ is continuous, then $f_\beta = \pi_\beta \circ f$ is also continuous. Indeed, regardless of the topology on $\prod_\alpha X_\alpha$, the function π_β is continuous. ■

Problem 19.7. *Let $\mathbb{R}^\infty$ be the subset of $\mathbb{R}^\omega$ consisting of all sequences that are "eventually zero," that is, all sequences $(x_1, x_2, \dots)$ such that $x_i \neq 0$ for only finitely many values of i . What is the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ in the box and product topologies? Justify your answer.*

Proof. We will show that $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$ in the product topology. Clearly $\overline{\mathbb{R}^\infty} \subseteq \mathbb{R}^\omega$. Let $(x_n) \in \mathbb{R}^\omega$. Let $U \subseteq \mathbb{R}^\omega$ be an open neighborhood (in the product topology) of (x_n) . Then $U = \prod_{n=1}^\infty U_n$, where $U_n \subseteq \mathbb{R}$ is open and $U_n = \mathbb{R}$ for all but finitely many n . Let $F = \{n \in \mathbb{N} \mid U_n \neq \mathbb{R}\}$. Let:

$$(y_n) = \begin{cases} x_n, & n \in F \\ 0, & n \notin F \end{cases}.$$

Then $(y_n) \in U$ and $(y_n) \in \mathbb{R}^\infty$. Hence $U \cap \mathbb{R}^\infty \neq \emptyset$, which means $(x_n) \in \overline{\mathbb{R}^\infty}$. Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$.

We will show that $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$ in the box topology. For any $(x_n) \notin \mathbb{R}^\infty$, then $x_n \neq 0$ for infinitely many n . Let $I = \{n \in \mathbb{N} \mid x_n \neq 0\}$. Consider the open set:

$$\prod_{n \in \mathbb{N}} U_n,$$

where $U_n = \mathbb{R} \setminus \{0\}$ for all $n \in I$ and $U_n = \mathbb{R}$ otherwise. Then $(x_n) \in U$ and $U \cap \mathbb{R}^\infty = \emptyset$. Hence $(x_n) \notin \overline{\mathbb{R}^\infty}$. Clearly $\mathbb{R}^\infty \subseteq \overline{\mathbb{R}^\infty}$. Conversely, we have just shown that if $(x_n) \notin \mathbb{R}^\infty$, then $(x_n) \notin \overline{\mathbb{R}^\infty}$, and hence $\overline{\mathbb{R}^\infty} \subseteq \mathbb{R}^\infty$. Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$. ■

Problem 23.10. Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space:

$$X = \prod_{\alpha \in J} X_\alpha.$$

Let (a_α) be a fixed point of X .

- (a) Given any finite subset K of J , let X_K denote the subspace of X consisting of all points (x_α) such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.
- (b) Show that the union Y of the spaces X_K is connected.
- (c) Show that X equals the closure of Y , conclude that X is connected.

Proof. (a) Let $K = \{\alpha_1, \dots, \alpha_n\}$. Define $\pi_K : X_K \rightarrow X_{\alpha_1} \times \dots \times X_{\alpha_n}$ by:

$$\pi_K((x_\alpha)) = (x_{\alpha_1}, \dots, x_{\alpha_n}).$$

This map is continuous, and its inverse given by:

$$(y_{\alpha_1}, \dots, y_{\alpha_n}) \mapsto \begin{cases} y_\alpha, & \alpha \in K \\ a_\alpha, & \alpha \notin K \end{cases}.$$

Hence π_K is a homeomorphism. Since K is finite and each X_α is connected, the finite product $X_{\alpha_1} \times \dots \times X_{\alpha_n}$ is connected. Therefore X_K is connected.

(b) Each X_K is connected and the fixed point (a_α) belongs to every X_K . The union of connected subspaces with a point in common is connected, establishing that Y is connected.

(c) Clearly $\bar{Y} \subseteq X$, so we only need to show the other direction containment. Let $(x_\alpha) \in X$. Let U be a neighborhood of (x_α) . Then $U = \prod_{\alpha \in J} U_\alpha$, where each $U_\alpha \subseteq X_\alpha$ is open, $x_\alpha \in U_\alpha$, and $U_\alpha = X_\alpha$ for all but finitely many α . Let:

$$F = \{\alpha \in J \mid U_\alpha \neq X_\alpha\}.$$

Define $(y_\alpha) \in X$ by:

$$y_\alpha = \begin{cases} x_\alpha, & \alpha \in F \\ a_\alpha, & \alpha \notin F. \end{cases}$$

Since $y_\alpha = a_\alpha$ for all $\alpha \notin F$, we have $(y_\alpha) \in Y$. Moreover, for all $\alpha \in F$, we have $y_\alpha = x_\alpha \in U_\alpha$. Likewise, for all $\alpha \notin F$, then $U_\alpha = X_\alpha$, hence $y_\alpha \in U_\alpha$. Thus $(y_\alpha) \in U$. This means $U \cap Y \neq \emptyset$, and since U was arbitrary we have $(y_\alpha) \in \bar{Y}$. Since the closure of a connected space is connected, we conclude that X is connected. ■

§ 2. The Urysohn Lemma

Problem 33.2(a). Show that a connected normal space having more than one point is uncountable.

Proof. Let X be connected and normal. Let x_0 and x_1 be distinct points in X . Since X is normal, $\{x_0\}$ and $\{x_1\}$ are closed in X . Moreover, $\{x_0\} \cap \{x_1\} = \emptyset$. By Urysohn's Lemma, there exists a continuous function $f : X \rightarrow [a, b]$ such that $f(x_0) = a$ and $f(x_1) = b$. Since f is continuous and X is connected, $f(X)$ is also connected. It must be the case that $f(X)$ is an interval—otherwise $f(X)$ contains only a single element, which is immediately a contradiction as x_0 and x_1 are distinct points which map to two distinct elements in $[a, b]$. Since $f(X)$ is an interval, it is uncountable, which implies X is uncountable. ■

Problem 33.3. Give a direct proof of the Urysohn lemma for a metric space (X, d) by setting:

$$f(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}.$$

Proof. Let X be a metric space. Let A and B be disjoint closed subsets of X . Since A and B are closed, $d(x, A) > 0$ for any $x \notin A$ and $d(x, B) > 0$ for any $x \notin B$. Moreover, $d(\cdot, A)$ and $d(\cdot, B)$ are continuous functions. Thus $f : X \rightarrow [0, 1]$ as given above is continuous, and $f(x) = 0$ for all $x \in A$ and $f(x) = 1$ for all $x \in B$. ■

Problem 33.8. Let X be completely regular, let K and C be disjoint closed subsets of X . Show that if K is compact, there is a continuous function $f : X \rightarrow [0, 1]$ such that $f(K) = \{0\}$ and $f(C) = \{1\}$.

Proof. Let $k \in K$. Then $k \notin C$. By the complete regularity of X , there exists $f'_k : X \rightarrow [0, 1]$ such that $f'_k(k) = 1$ and $f'_k(C) = \{0\}$. Define $f_k := 1 - f'_k$. Then $f_k(k) = 0$ and $f_k(C) = \{1\}$. Note that $[0, \frac{1}{2})$ is open in $[0, 1]$, hence $f_k^{-1}([0, \frac{1}{2}))$ is open in X . Moreover, it is clear that $k \in f_k^{-1}([0, \frac{1}{2}))$. We've obtained a cover $K \subseteq \bigcup_{k \in K} f_k^{-1}([0, \frac{1}{2}))$. Since K is compact, there exists points $\{k_1, \dots, k_n\}$ such that $K \subseteq \bigcup_{i=1}^n f_{k_i}^{-1}([0, \frac{1}{2}))$. Define $f := \min\{f_{k_1}, \dots, f_{k_n}\}$. Then $f(K) = [0, \frac{1}{2})$ and $f(C) = \{1\}$. Define $\varphi : [0, 1] \rightarrow [0, 1]$ by:

$$\varphi(t) := \begin{cases} 0, & 0 \leq t < \frac{1}{2} \\ 1, & t = 1 \end{cases}.$$

Since φ is continuous, the composition $\varphi \circ f$ is continuous. In particular, we have $(\varphi \circ f)(K) = \varphi([0, \frac{1}{2})) = \{0\}$ and $(\varphi \circ f)(C) = \varphi(\{1\}) = \{1\}$. This establishes the claim. ■

§ 3. The Baire Category Theorem

Problem 3.1. Show that if $\bigcup B_n$ is a nonempty Baire space, at least one of the sets $\overline{B_n}$ has a nonempty interior.

Proof. Suppose towards contradiction $\overline{B_n}^\circ = \emptyset$ for each n . Consider the collection of closed sets $\{\overline{B_n}\}_n$. Since $\bigcup B_n$ is a Baire space, it must be that $\left(\bigcup \overline{B_n}\right)^\circ = \emptyset$. But this means:

$$\begin{aligned} \bigcup B_n &\subseteq \bigcup \overline{B_n}^\circ \\ &\subseteq \left(\bigcup \overline{B_n}\right)^\circ \\ &= \emptyset. \end{aligned}$$

Thus $\bigcup B_n = \emptyset$, which is a contradiction. Therefore at least one of the sets $\overline{B_n}$ has a nonempty interior. ■

Problem 3.2. The Baire Category Theorem implies that $\mathbb{R}$ cannot be written as the countable union of closed subsets having empty interiors. Show this fails if the sets are not required to be closed.

Proof. Let $\{q_i\}_{i=1}^\infty$ be an enumeration of the rationals. Then $\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q} = (\mathbb{R} \setminus \mathbb{Q}) \cup \bigcup_{i=1}^\infty \{q_i\}$. Each singleton $\{q_i\}$ has empty interior, and $\mathbb{R} \setminus \mathbb{Q}$ also has an empty interior. Hence $\mathbb{R}$ is a countable union of subsets with empty interior, so the conclusion fails without the assumption that the sets are closed. ■

Problem 3.3. Show that the irrationals are a Baire space.

Proof. Let $\{U_n\}_{n=1}^\infty$ be a sequence of dense open subsets of $\mathbb{R} \setminus \mathbb{Q}$. For each n , we can write $U_n = V_n \cap \mathbb{R} \setminus \mathbb{Q}$, where V_n is open in $\mathbb{R}$. Since $V_n \cap \mathbb{R} \setminus \mathbb{Q}$ is dense, and because $V_n \cap \mathbb{R} \setminus \mathbb{Q} \subseteq V_n$, we can conclude that V_n is dense. Now if $\{q_i\}_{i=1}^\infty$ is an enumeration of the rationals, we have:

$$\begin{aligned} \bigcap_{n=1}^\infty (V_n \cap \mathbb{R} \setminus \mathbb{Q}) &= \left(\bigcap_{n=1}^\infty V_n \right) \cap \mathbb{R} \setminus \mathbb{Q} \\ &= \left(\bigcap_{n=1}^\infty V_n \right) \cap \left(\bigcap_{n=1}^\infty \{q_n\}^c \right). \end{aligned}$$

Since $\mathbb{R}$ is complete, by the Baire Category Theorem it is a Baire space. Since $\left(\bigcap_{n=1}^\infty V_n\right) \cap \left(\bigcap_{n=1}^\infty \{q_n\}^c\right)$ is a countable intersection of open and dense subsets of $\mathbb{R}$, we conclude that $\left(\bigcap_{n=1}^\infty V_n\right) \cap \left(\bigcap_{n=1}^\infty \{q_n\}^c\right)$ is dense. Thus $\bigcap_{n=1}^\infty U_n$ is dense, establishing that $\mathbb{R} \setminus \mathbb{Q}$ is Baire. ■

Problem 1. Let (M, d) be a complete metric space with no isolated points. Prove that M is uncountable. (A point x in a topological space X is called a isolated point if the one point set $\{x\}$ is open in X).

Proof. Suppose towards contradiction M is countable. Since M has no isolated points, for each $x \in M$ the set $\{x\}^c$ is open. Moreover, $\overline{\{x\}^c} = (\{x\}^o)^c = \emptyset^c = M$. Hence $\{x\}$ is open and dense in M . But:

$$\begin{aligned} \bigcap_{i=1}^{\infty} \{x_i\}^c &= \left(\bigcup_{i=1}^{\infty} \{x_i\} \right)^c \\ &= M^c \\ &= \emptyset. \end{aligned}$$

Since we've found a countable collection of open and dense sets whose countable intersection is *not* dense, M is not a Baire Space. This contradicts the Baire Category Theorem, as M being a complete metric spaces implies that it is a Baire space. It must be the case that M is uncountable. ■

Problem 2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function, and suppose that for each $x \in \mathbb{R}$, there exists a natural number n such that $f^{(n)}(x) = 0$. For each fixed $n \in \mathbb{N}$, consider the set $F_n := \{x \in \mathbb{R} \mid f^{(n)}(x) = 0\}$. Let $[a, b]$ be any closed interval.

(a) Show that $[a, b] = \bigcup_{n=0}^{\infty} ([a, b] \cap F_n)$.

(b) Show that there is a nonempty open interval $A \subseteq [a, b]$ where f is a polynomial.

Proof. (a) Consider the set $\bigcup_{n=0}^{\infty} F_n$. Clearly $\bigcup_{n=0}^{\infty} F_n \subseteq \mathbb{R}$. If $x \in \mathbb{R}$, then there exists $i \in \mathbb{N}$ such that $f^{(i)} = 0$, hence $x \in F_i$. Thus $x \in \bigcup_{n=0}^{\infty} F_n$, which shows $\mathbb{R} = \bigcup_{n=0}^{\infty} F_n$. With this:

$$\begin{aligned} \bigcup_{n=0}^{\infty} ([a, b] \cap F_n) &= [a, b] \cap \bigcup_{n=0}^{\infty} F_n \\ &= [a, b] \cap \mathbb{R} \\ &= [a, b]. \end{aligned}$$

(b) Since $[a, b]$ is a compact Hausdorff space, $[a, b]$ is a Baire space. By part (a), this implies $\bigcup_{n=0}^{\infty} ([a, b] \cap F_n)$ is a Baire space. Note that $F_n = \{x \in \mathbb{R} \mid f^{(n)}(x) = 0\} = (f^{(n)})^{-1}(\{0\})$. Thus F_n is closed, which means $[a, b] \cap F_n$ is closed. By Problem 3.1, there exists some $i \in \mathbb{N}$ such that $[a, b] \cap F_n$ has nonempty interior. Therefore there exists some open interval $A \subseteq [a, b] \cap F_n$. Since $A \subseteq F_n$, for all $a \in A$ we have $f^{(i)}(a) = 0$. Therefore f is a polynomial on $A \subseteq [a, b]$. ■